

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: IM Industrial and Systems Engineering

2. Subject Name: Work Systems and Ergonomics
Credits: 3

L-T-P: 3-0-0

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Work Systems:

Industry 1.0, 2.0, 3.0, 4.0 and 5.0 work systems; Major Changes. Work system performance issues: productivity, quality, safety, reliability, ergonomics. Manual, mechanised and automated work systems. Technologies for designing work systems: Virtual and Augmented Reality, Eye Tracker, Internet of Things (IoT), 3D Motion Capture Systems;

Fundamentals of Work System Design:

Method Study: Exploratory Tools, Recording and Analysis Tools, Quantitative Tools, Worker and Machine Relationships. Work Measurement: Work Sampling, Direct Time Study, Concepts of Rating and Allowances, Setting Standard Times for Jobs, Standard Data, Predetermined Time Standards: Work-Factor, and Methods-Time-Measurement. Anthropometric Principles and Postural Analysis in Workspace Design: Anthropometry and its uses, principles of applied anthropometry, applications of anthropometry in design, postures and body mechanics, musculoskeletal problems in sitting and standing. Design of Physical Environment: Human thermoregulation, measuring thermal environment, measurement of light, lighting design considerations, measurement of sound, industrial noise control, vibration.

Human –Machine Interaction:

Design of Interactive Systems: Human, Machine, Environment and their Interaction. Heuristics for Interaction Design and Evaluation Techniques. Human Information Processing Model: Perception and Signal Detection Theory, Memory, Decision Making and Response Selection, Response Execution, Attention Resources and Multitasking. Cognitive Models in HCI: GOMS family of models (KLM and CMN-GOMS), Motor Behaviour Models (Fitts' law and Hick Hyman's law). Task Modelling and Analysis: Hierarchical task analysis (HTA), Engineering task models and Concur Task Tree (CTT). Design Principles: Coding of Information, Display of Visual and Auditory Information. Smartphones, tablets, Virtual Reality (VR) and Augmented Reality (AR) gears, voices, and gesture.

Human-Machine-Autonomy interaction:

Cyber physical system architectures in Smart manufacturing; Systems, Instruments and Alarms, Software. Design principles of Virtual Reality-based robot interface, Model-Mediated teleoperation (MMT), human-robot collaboration (HRC), and tactile internet.

Reference Books:

- 1) Niebel's Methods, Standards, and Work Design (12th Ed.), Freivalds, A., & Niebel, B. W., McGraw Hill, 2013
- 2) Work Systems: The Methods, Measurement and Management of Work, Groover, M. P., Pearson, 2014
- 3) Engineering Psychology and Human Performance (4th Ed.), Wickens, C. D., Hollands, J. G., Banbury, S., & Parasuraman, R., Pearson, 2013
- 4) Human Computer Interaction (3rd Ed.), Dix, A., Finlay, J., Abowd, G. D., & Beale, R., Pearson, 2004

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Work Systems: Industry 1.0, 2.0, 3.0, 4.0 and 5.0 work systems; Major Changes. Work system performance issues: productivity, quality, safety, reliability, ergonomics. Manual, mechanised and automated work systems. Technologies for designing work systems: Virtual and Augmented Reality, Eye Tracker, Internet of Things (IoT), 3D Motion Capture Systems;	8
2.	Fundamentals of Work System Design: Method Study: Exploratory Tools, Recording and Analysis Tools, Quantitative Tools, Worker and Machine Relationships. Work Measurement: Work Sampling, Direct Time Study, Concepts of Rating and Allowances, Setting Standard Times for Jobs, Standard Data, Predetermined Time Standards: Work- Factor, and Methods-Time-Measurem ent. Anthropometric Principles and Postural Analysis in Workspace Design: Anthropometry and its uses, principles of applied anthropometry, applications of anthropometry in design, postures and body mechanics, musculoskeletal problems in sitting and standing. Design of Physical Environment: Human thermoregulation, measuring thermal environment, measurement of light, lighting design considerations, measurement of sound, industrial noise control, vibration.	12
3.	Human –Machine Interaction: Design of Interactive Systems: Human, Machine,	

	Environment and their Interaction. Heuristics for Interaction Design and Evaluation Techniques. Human Information Processing Model: Perception and Signal Detection Theory, Memory, Decision Making and Response Selection, Response Execution, Attention Resources and Multitasking. Cognitive Models in HCI: GOMS family of models (KLM and CMN-GOMS), Motor Behaviour Models (Fitts' law and Hick Hyman's law). Task Modelling and Analysis: Hierarchical task analysis (HTA), Engineering task models and Concur Task Tree (CTT). Design Principles: Coding of Information, Display of Visual and Auditory Information. Smartphones, tablets, Virtual Reality (VR) and Augmented Reality (AR) gears, voices, and gesture.	12
4.	Human-Machine-Autonomy interaction: Cyber physical system architectures in Smart manufacturing; Systems, Instruments and Alarms, Software. Design principles of Virtual Reality-based robot interface, Model-Mediated teleoperation (MMT), human-robot collaboration (HRC), and tactile internet.	4
Total number of hours		36